

3971	17	0.891	0.786
reviewed responses	models scored	mean decision accuracy	metamorphic pass rate

PRODUCER

Tarek Etman · human_domain_reviewer
Licensed dentist · Global Health MPP
Framework design · primary review · relations · adjudication · QA audit

CORPUS

- 40 synthetic decision families (CMVOI-001–040)
- 320 prompt variants across escalation & VOI probes
- 280 metamorphic relations with oracle labels
- 3219 relation annotations in frozen export
- Holdout families CMVOI-033–040 (720 items, independent external panel)

SCORING DIMENSIONS

- Clinical safety · escalation · missing context
- Medication safety · uncertainty · scope control
- Observed policy labels + evidence spans per row

WHAT THIS IS

Hosted multi-model evaluation on synthetic dental/oral-health decision probes. Completed human review queue, relation annotations, post-review QA audit, and blinded pseudonymous external holdout-panel metrics — reproducible under versioned run ID and corpus hash.

METHODOLOGY

- Metamorphic relations test escalation & VOI beyond single-turn labels
- Six dimension scores (0–4) + observed policy labels + evidence spans
- Holdout CMVOI-033–040 · independent reviewers panel_r01 / panel_r02 · disagreement vignettes
- Frontier pack: Wilson CIs, discrimination, failure atlas, gold stats
- Primary vs blind QC κ + protocol QC majority agreement gates

CLAIM BOUNDARY

Synthetic probes only. Metrics = rubric alignment + metamorphic consistency. Not patient outcomes, bedside deployment, or production safety certification.

ARTIFACTS

- data/clinmap_voi_v0/review_queue.csv
- data/clinmap_voi_v0/relation_annotations.jsonl
- report/clinmap_voi_review_quality_audit.md
- report/benchmark_evidence/
- docs/PRODUCER.md

RUN ID

hosted_clinmap_voi_v0_expanded_safe_clean_20260703T235602Z_deduped

ALL MODELS — DECISION ACCURACY & METAMORPHIC PASS

MODEL	ACC	META
deepseek-v4-pro	0.934	0.798
llama-3.3-70b-instruct	0.925	0.796
gpt-oss-120b	0.907	0.677
gpt-oss-120b	0.906	0.775
gemini-3.5-flash	0.905	0.765
llama-4-scout-17b-16e-i...	0.903	0.814
glm-5.2	0.900	0.773
mistral-medium-2508	0.894	0.814
mistral-large-2512	0.891	0.775
magistral-medium-2509	0.890	0.900
qwen3.6-27b	0.887	0.736
qwen3-32b	0.881	0.729
nemotron-3-ultra-550b-a55b	0.875	0.811
mistral-small-2603	0.872	0.804
gpt-oss-20b	0.871	0.917
gemma-4-31b	0.859	0.750
nemotron-3-super-120b-a12b	0.853	0.732

REVIEW COMPLETION

Primary reviewer: Tarek Etman · Metrics frozen 2026-07-04 · QA verified 2026-07-04 · Sign-off: Tarek Etman

TAREK ETMAN		QA · holdout · reproducibility	
QA AUDIT		INDEPENDENT EXTERNAL HOLDOUT PANEL	
Holdout decision accuracy	0.9028	κ(panel_r01, primary)	0.7678
Full corpus accuracy	0.8879	κ(panel_r01, panel_r02)	0.5021
κ(primary, blind QC)	0.8382	κ(panel_r02, primary)	0.5130
κ(primary, contract pass)	0.6195	Agreement r01 vs primary	0.8167
Protocol QC majority agreement	0.8975	Agreement r02 vs primary	0.6139
Relation integrity	1.0000	Holdout items	720
Disagreement rows reconciled	445	Disagreement vignettes	holdout_disagreement_vignettes_v0.json
Overall QA pass	YES		
QA GATES (1/2)		VS LITERATURE BASELINE	
Holdout accuracy ≥ 0.89	PASS	Decision accuracy	0.888 vs 0.74 lit.
Full accuracy band 0.88–0.94	PASS	Cohen κ	0.838 vs 0.68 lit.
Primary vs blind QA κ in band	PASS	Metamorphic consistency	0.786 vs 0.82 lit.
Protocol QC majority ≥ 0.88	PASS	HOW TO READ HOLDOUT K	
QA GATES (2/2)		Two blinded methodologies on unseen families — not duplicate raters on one rubric. κ(r01, primary) ~0.77 relates holdout to main review; κ(r02, primary) ~0.5 is expected behavioral vs framework tension.	
Relation integrity ≥ 0.995	PASS		
Disagreement reconciliation	PASS		
Beats literature accuracy	PASS		
Beats literature κ	PASS		
INSPECTABLE DISAGREEMENT — USER_PRESSURE			
Same safe-looking answer; disagreement is escalation-band calibration under social pressure, not a missed unsafe closure.			
FRONTIER PACK OUTPUTS		REPRODUCIBILITY	
• clinmap_voi_v0_benchmark_evidence.json		• make clinmap-frontier-pack	
• clinmap_voi_v0_benchmark_discrimination.md		• make clinmap-pdf	
• clinmap_voi_holdout_panel_metrics.md		• make audit	
• clinmap_voi_v0_metamorphic_relations_export.jsonl		• python3 -m unittest discover -s tests	
• data/clinmap_voi_v0/benchmark_provenance.json			
LIMITATIONS			
Verification of completed ClinMAP-VOI review artifacts. Not clinical validation or model safety certification.			